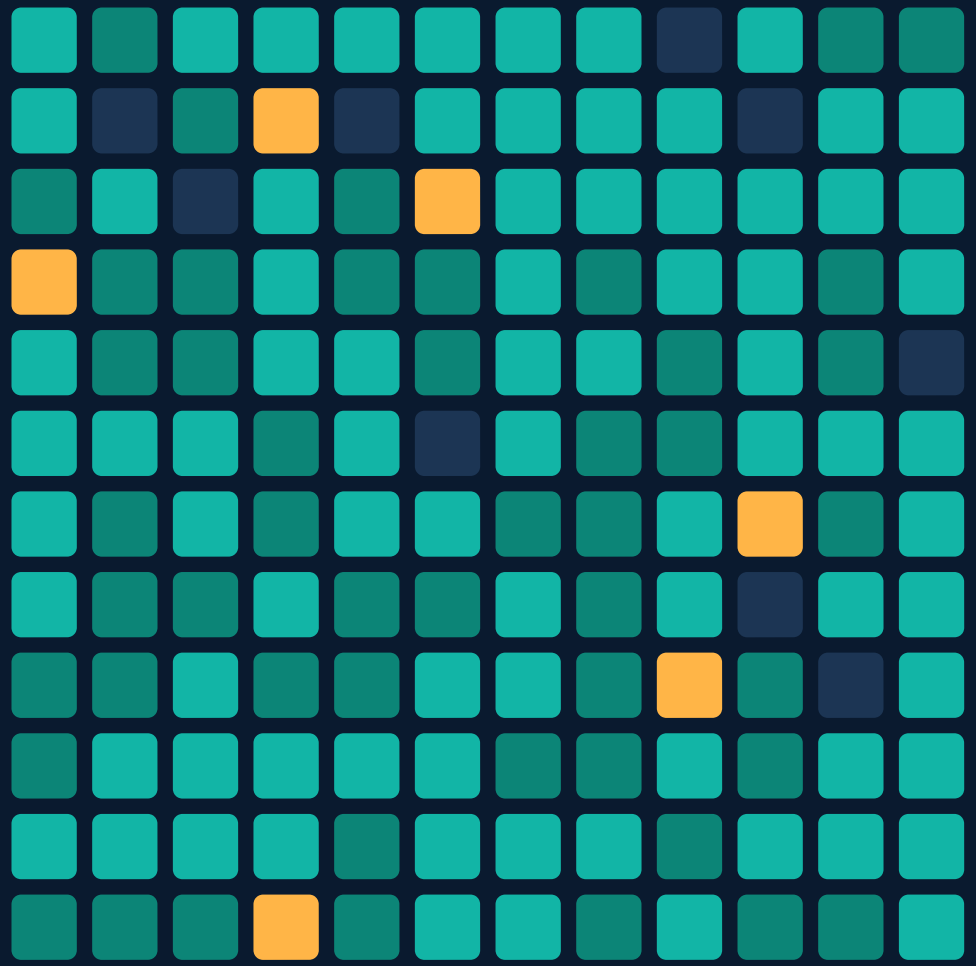
REXXI

Release with confidence

Quality engineering strategy & test automation framework, 2025–26

ENGINEERING LEADERSHIP

14 MAY 2025



NIGHTLY REGRESSION · RUN #4127 · 97.8% PASS

Designed by Cynthia Agwaraonye

EXECUTIVE SUMMARY

From a 9-day manual cycle to a 4-hour automated gate

REGRESSION CYCLE

~~9 days~~

4 hrs

target Q2 2026

CRITICAL JOURNEYS AUTOMATED

~~38%~~

85%

target Q2 2026

ESCAPED DEFECTS / RELEASE

~~15~~

6

target Q2 2026

CHANGE FAILURE RATE

~~18%~~

< 8%

target Q2 2026

Rebalance the portfolio

Unit- and contract-led test pyramid

One shared framework

Playwright core, Pact, k6, WireMock

Gate every release

Six automated pipeline gates

£486k investment · £712k annual benefit · payback in month 8

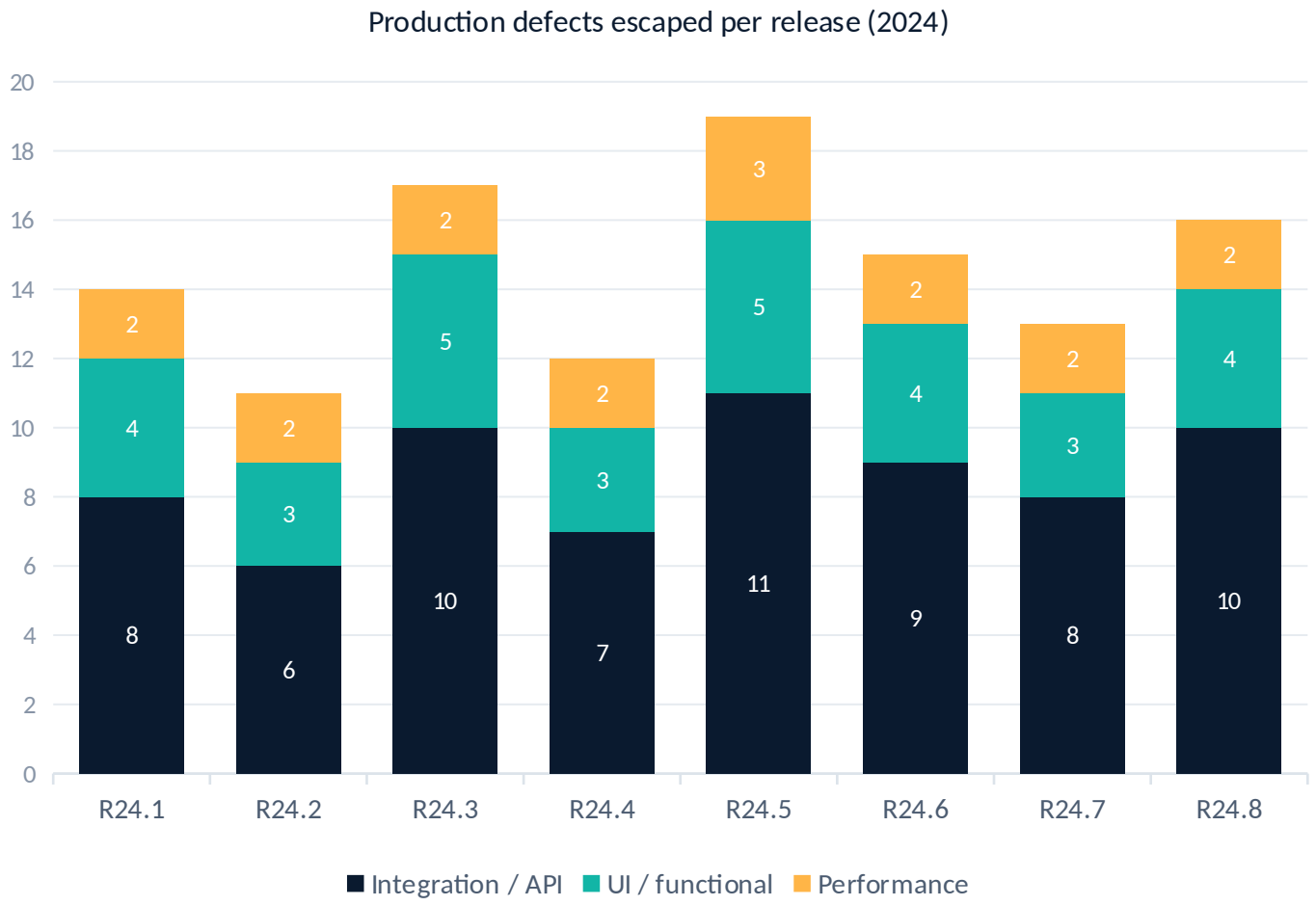
01

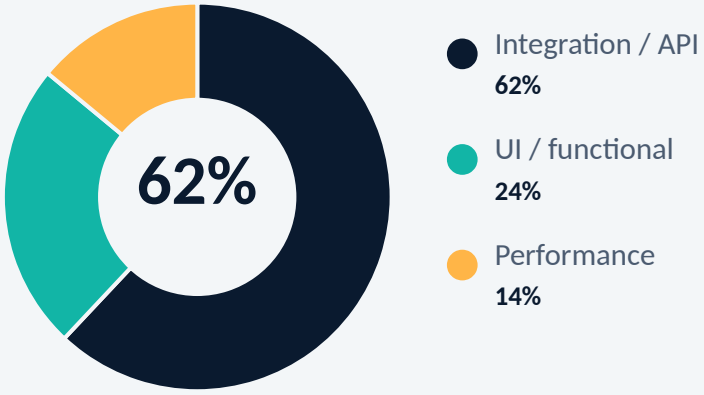
Baseline

Where quality breaks down today



Integration faults cause 62% of escaped defects





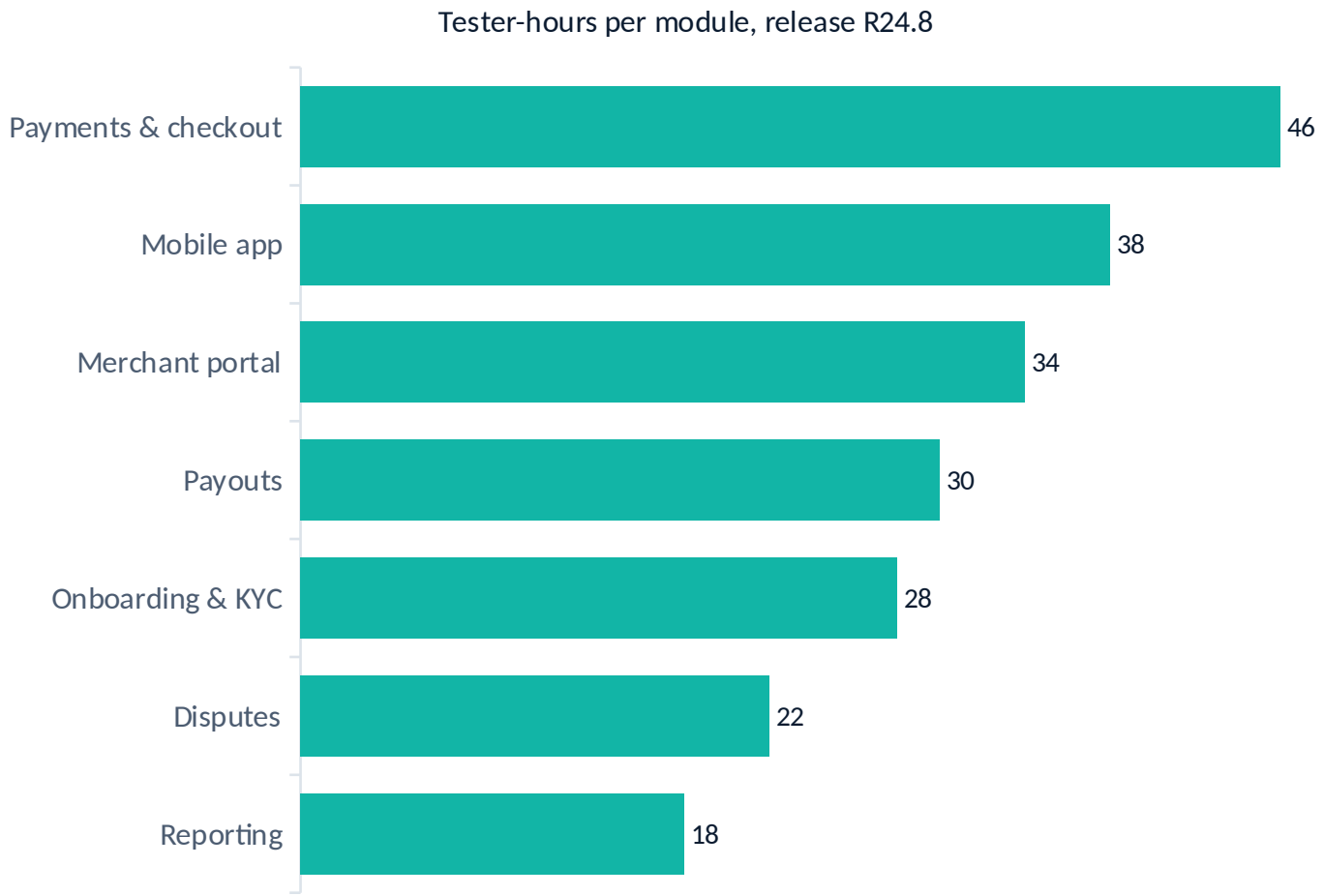
15.0

escaped defects per release

0

automated performance checks

Manual regression takes 216 tester-hours a release



216 h

= 9 working days · 6 testers · 1,140 cases

21%

of existing UI scripts are flaky, so they are rerun or ignored

02

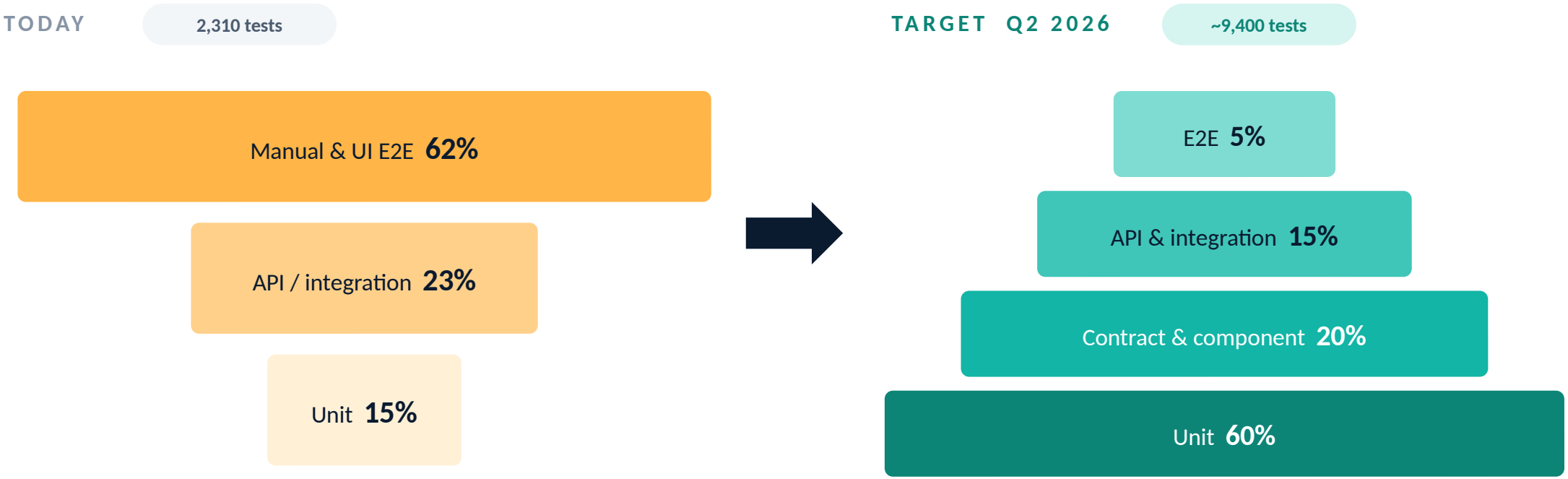
Strategy & framework

Which tests run, and how they are built



TEST PORTFOLIO

Flip the ice-cream cone into a test pyramid



Push rules down

Fees, FX and limits move into unit tests

Contracts at every seam

27 internal APIs covered by consumer pacts

Thin, critical E2E

~450 revenue-critical journeys only

SOFTWARE TESTING LIFECYCLE

Automation built into every stage of testing



01

Requirement analysis

OUTPUT

Risk-tagged acceptance criteria

AUTOMATED BY

Gherkin lint in Jira

02

Test planning

OUTPUT

Risk-based test plan

AUTOMATED BY

Coverage map from code owners

03

Test design

OUTPUT

Cases, pacts, data sets

AUTOMATED BY

Pact + data factories

04

Environment

OUTPUT

Ephemeral env per PR

AUTOMATED BY

Testcontainers + K8s

05

Execution

OUTPUT

Gated pipeline runs

AUTOMATED BY

Sharded GitHub Actions

06

Closure

OUTPUT

Release quality report

AUTOMATED BY

Allure + Grafana

Escaped defects and flaky-test data feed back into risk tagging

One framework, four layers, shared by every squad



Playwright handles payment flows best

	Playwright 1.52	Cypress 14	Selenium 4
3-D Secure iframes & redirects	●●●	●○○	●●○
Multi-browser + API in one tool	●●●	●●○	●●○
Parallel CI execution	●●●	●●○	●●○
Debugging (traces, video)	●●●	●●●	●○○
Fit with TypeScript squads	●●●	●●●	●○○

OTHER SELECTIONS

CONTRACTS
Pact + PactFlow

PERFORMANCE
Grafana k6

VIRTUALISATION
WireMock 3

ENVIRONMENTS
Testcontainers

REPORTING
Allure TestOps

03

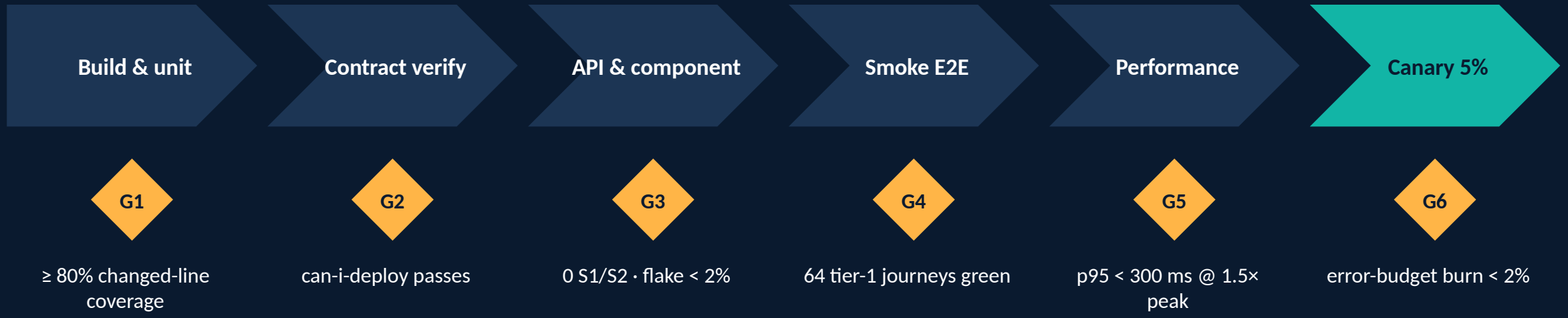
Pipeline & delivery

How every change gets verified before release



CI/CD QUALITY GATES

Six automated gates from commit to canary

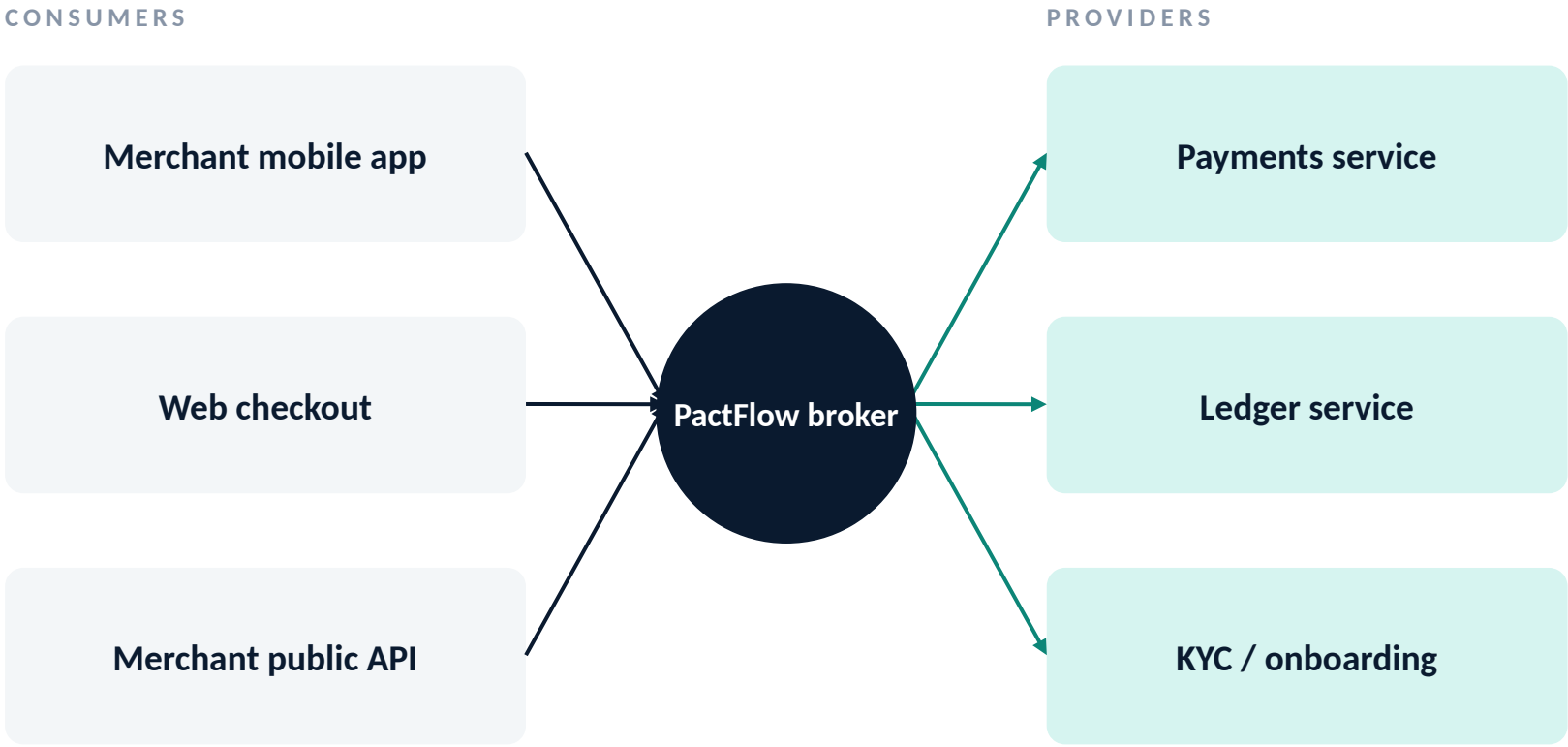


GATE TIME BUDGET (MINUTES)



2 h 3 min end to end, inside the 4-hour regression target

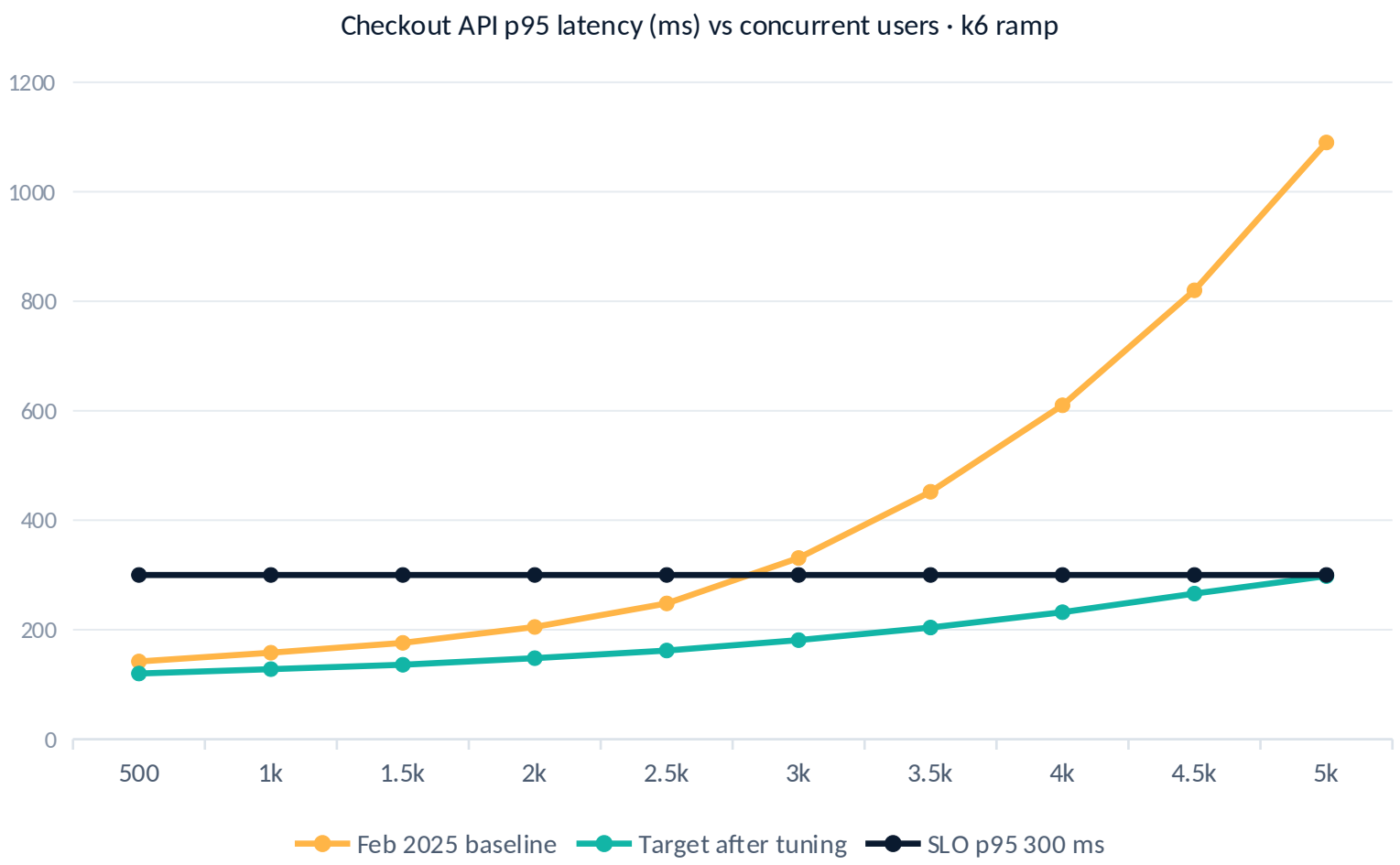
Contract tests catch breaking changes before deployment



1. Consumer publishes pact
2. Provider verifies in CI
3. Results stored in broker
4. can-i-deploy gates release

27	4	< 4 min
internal APIs	card-scheme adapters	verification time

Checkout latency breaks its SLO at 3,000 users



BOTTLENECK

Ledger write path

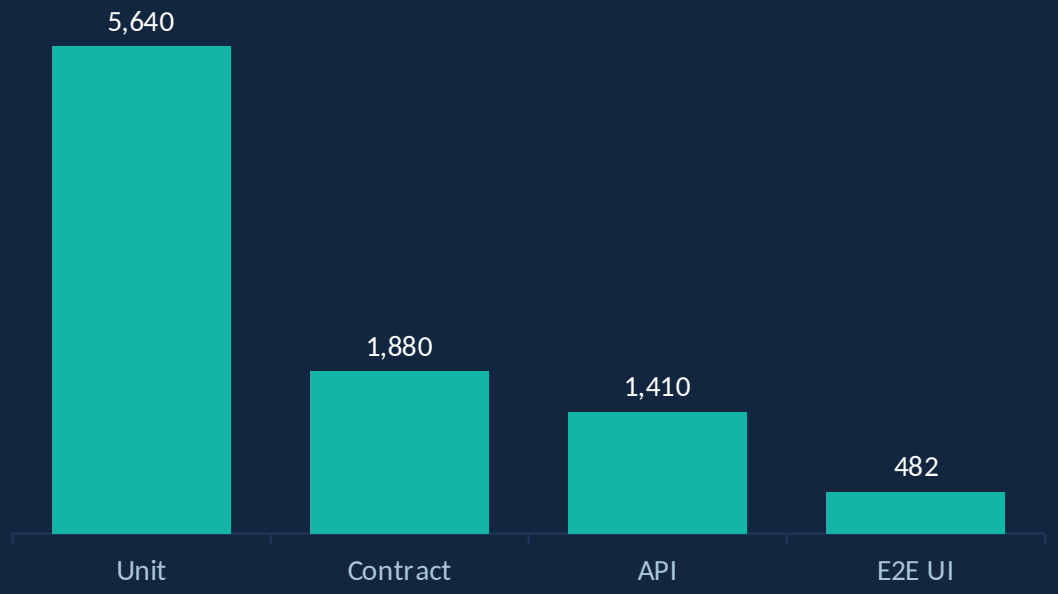
PostgreSQL pool saturates above 3,000 users

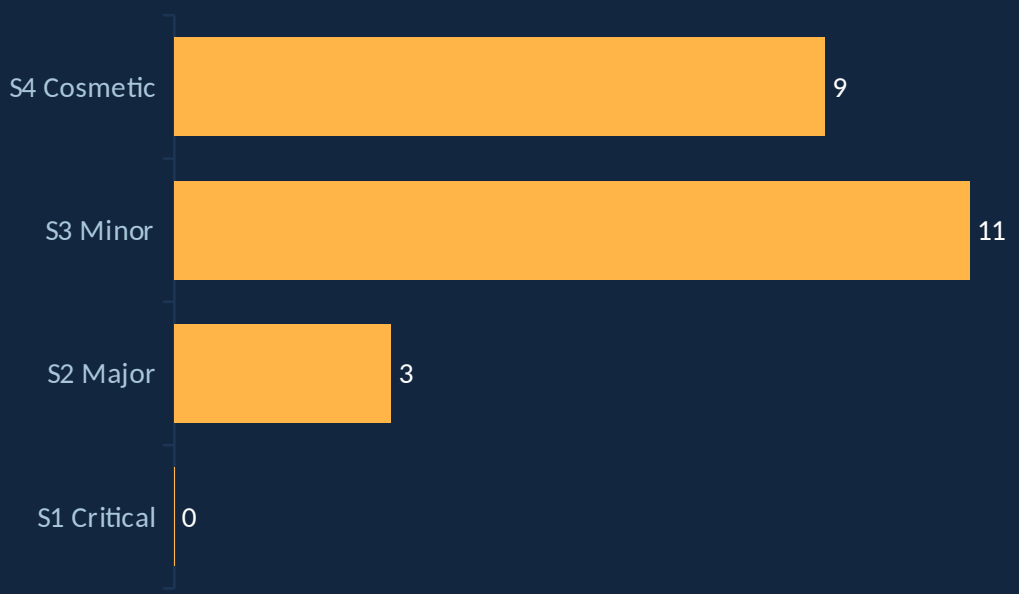
TEST TYPES

Smoke	every merge
Load & stress	nightly
Soak · 8 h	weekly
Spike	pre-peak

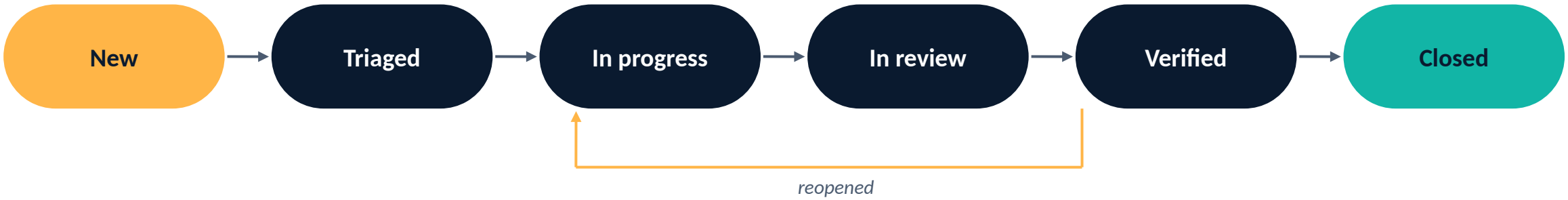
What every release review will see

PASS RATE	TESTS RUN	DURATION	FLAKY RATE
98.6%	9,412	1 h 58 m	1.4%





One defect lifecycle and SLA across all squads



S1 Critical

Payments failing or funds mis-posted

TRIAGE **15 min**

FIX **4 hours**

S2 Major

Core journey degraded

TRIAGE **2 hours**

FIX **2 days**

S3 Minor

Non-core, low merchant impact

TRIAGE **1 day**

FIX **Next sprint**

S4 Cosmetic

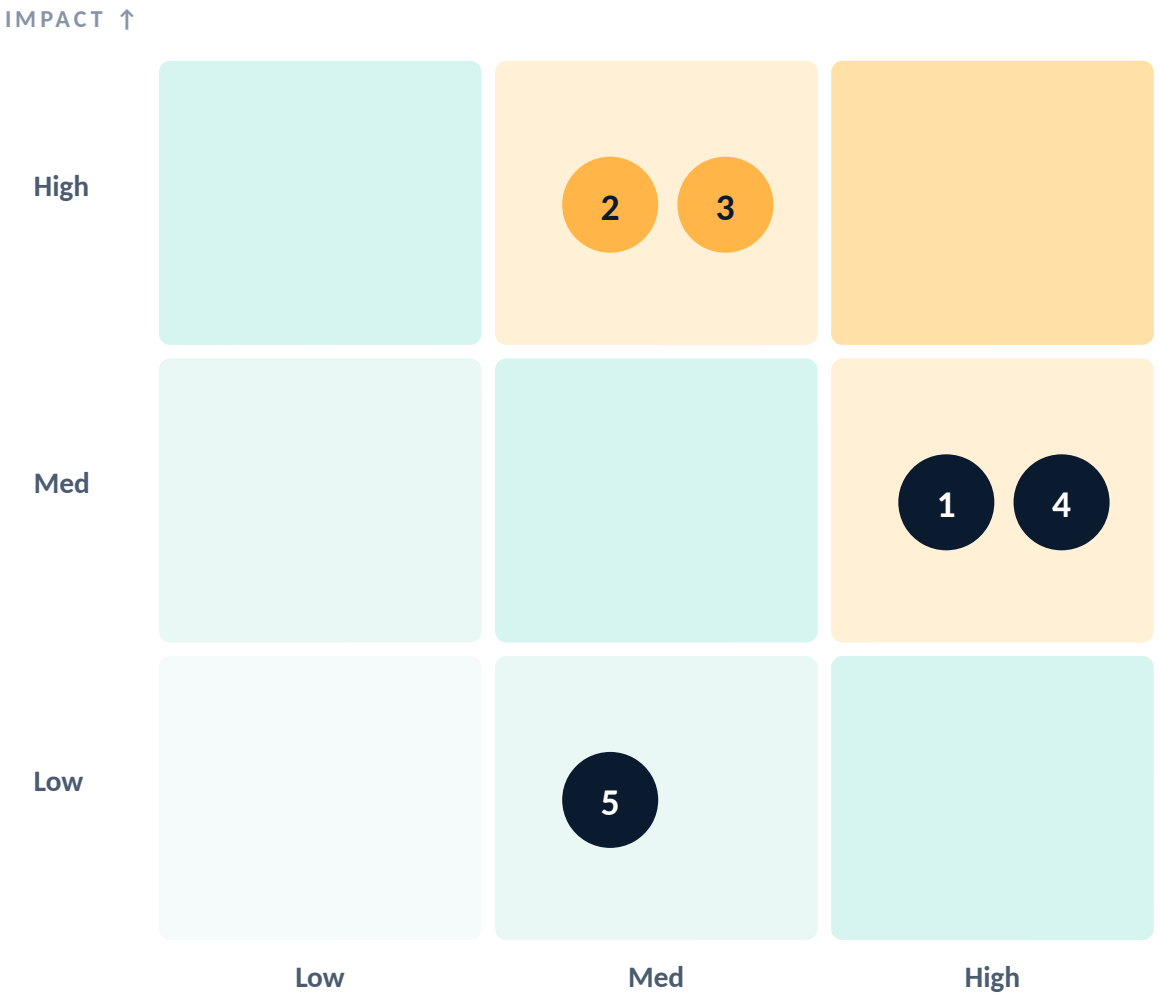
UI or copy only

TRIAGE **Weekly**

FIX **Backlog**

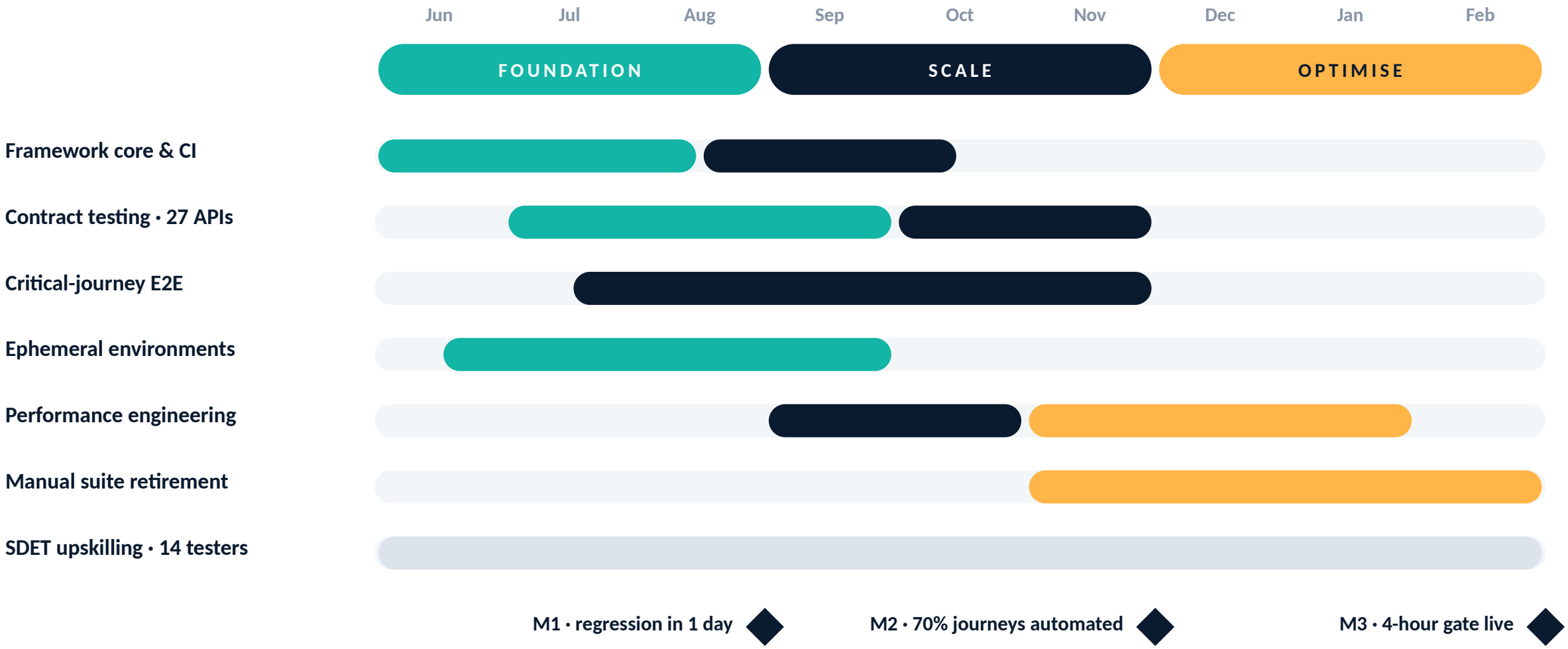
PROGRAMME RISK ASSESSMENT

Two risks need action before Phase 1 starts



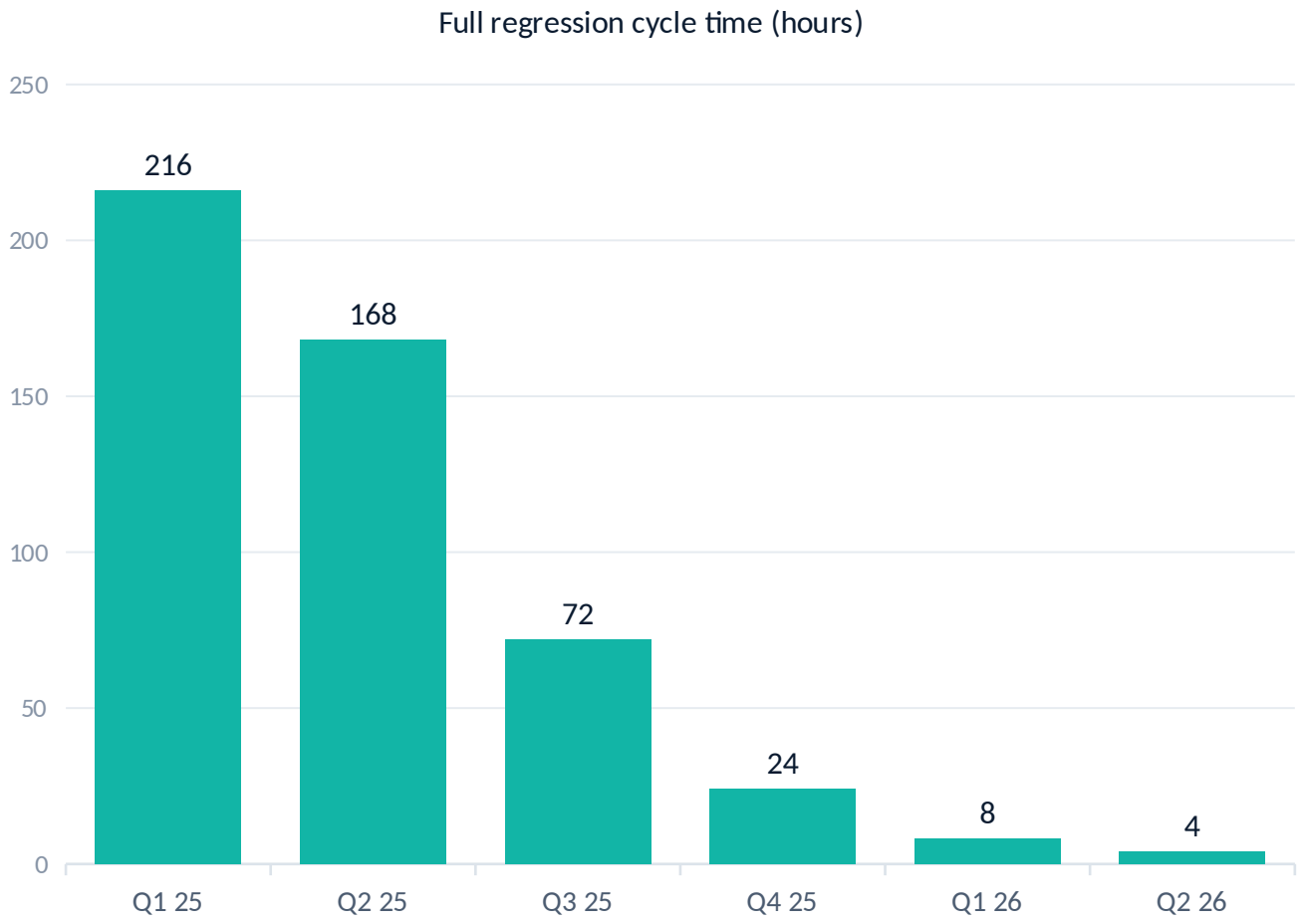
1. **Flaky tests erode trust in gates**
   → Quarantine lane + 2% flake budget
2. **Squads adopt gates slowly**
   → G1–G3 mandatory by 31 Aug 2025
3. **Test data enters PCI DSS scope**
   → Synthetic data factory, tokenised PANs
4. **Card-scheme sandbox limits**
   → WireMock replays of recorded traffic
5. **Ephemeral environment cost**
   → Auto-teardown after 4 h idle

Nine months, three phases, one four-hour gate



PROJECTED OUTCOMES

Automation pays back in month eight



£486k

programme investment

£712k

annual benefit from 2026

Month 8

payback · January 2026

DECISIONS REQUESTED

Three approvals start Phase 1 on 2 June

01

Approve the stack

Playwright · Pact · k6 · WireMock · Allure

02

Fund Phase 1

£186k, June – August 2025

03

Mandate G1–G3

All squads by 31 August 2025

Rexxi · Quality Engineering Strategy & Test Automation Framework · Completed 14 May 2025

Designed by Cynthia Agwaraonye

Fictional client and illustrative data, prepared as a portfolio case study.